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Estimation of rebound position in basketball

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Abstract

Basketball rebound regions are estimated using a dynamic model for short-, medium-, and long-range direct and bank shots and with previously measured standard deviations of release velocity and angle. The model includes basketball stiffness and damping, and contains flight and ball-contact sub-models. The deterministic simulations, together with random Gaussian probability density functions for release velocity, angle and lateral deviation angle, analyze ball trajectories from different shot positions. The results instruct likely rebounding positions. Angled direct and bank shots from the same release position usually have two high-probability rebound positions, one on the same side as and one on the opposite side from the shooter. The highest probability rebound position is typically in or near the optimal shot path plane for capture. Rebound positions from skilled player's shots are usually more predictable.

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1. Introduction

Excellent basketball rebounders have height, leaping ability including a sense of timing, and position placement skill. They seem to know where the ball will bounce off the rim or backboard. Prediction of rebound position plays a large role in their rebounding success. However, this prediction is difficult because the ball sometimes behaves chaotically around the rim and backboard.

Some previous studies provided recommendations for good rebounding positions with observations or measurements in actual basketball games and even based on experiments [1-4]. However, their results and conclusions were varied because they investigated different games, sets of player skills, and situations. Another more deliberate and controlled method to estimate the ball rebound position uses a simulation

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model. Okubo and Hubbard have calculated typical missed ball trajectories in free throws [5]. They estimated the rebound positions for field shots with an assumed model of release conditions [6].

No one has yet investigated the relationship between measured release conditions and ball rebound position. We here use a deterministic dynamic model to calculate ball trajectories. Then random initial conditions with standard deviations of release velocity and angle are based on the measured data of Toyoshima et al. [7]. We estimate high likelihood rebound positions for short-, medium- and long-range direct and bank shots using the previously measured release conditions of [7].

2. Method

2.1. Dynamic model

Our overall dynamic model for general basketball field shots has six distinct sub-models: gravitational flight with air drag, and ball-contact sub-models for ball-rim, ball-backboard, ball-bridge (hoop-mounting plate), ball-bridge-board and ball-rim-board contacts. We switch between the sub-models depending on the reaction forces at the ball-contact points. The flight sub-model is adopted when the normal force at each contact point vanishes. Each contact sub-model has possible slipping and non-slipping, and spinning and non-spinning motions. The equations of motion have been derived for the flight and ball-contact sub-models including equivalent ball contact stiffness and damping.

2.2. Simulation parameters

A Newtonian frame with origin at the hoop center has its *XY* plane horizontal, the direction of *Y* from the hoop center toward the board, and *Z* up. Ball release location is specified by three parameters: horizontal and vertical distances l and h, from the hoop center, and the floor angle β, between the board surface and the vertical plane including the release point and hoop center. Shots have four more release parameters: release velocity v, backspin angular velocity ω, release angle α, between the release velocity and the horizontal plane, and lateral deviation angle δ, between the initial ball path plane and the vertical plane including the hoop center and the release point. We choose direct and bank shots with short ($l = 2$ m, $h = 0.05$ m, $\omega = 1$ Hz), medium ($l = 4$ m, $h = 0.15$ m, $\omega = 2$ Hz), and long ($l = 7$ m, $h = 0.30$ m, $\omega = 3$ Hz) ranges. Direct shots are calculated at seven different floor angles: 0 (near the baseline), 15°, 30°, 45°, 60°, 75° and 90° (in the board normal plane including the hoop center), and bank shots at six floor angles: 15°, 30°, 45°, 60°, 75° and 90°. All shots are taken from the right side of the court. The simulations use as initial conditions the release velocity, angle and lateral deviation angle in increments of 0.02 m/s, 0.5°, and 0.5°, respectively, for short-range shots; 0.02 m/s, 0.5°, and 0.25° for medium-range shots; and 0.02 m/s, 0.5°, and 0.15° for long-range shots.

2.3. Experimental data and assumption

We assume that shooters choose an optimal aiming point for release velocity, angle and lateral deviation angle $(\bar{v}, \bar{\alpha}, \bar{\delta})$. The aiming point allows the largest total error margins in the three directions $(M_v, M_\alpha, M_\delta)$ for capture. For the three parameters, we use shots chosen from independent Gaussian probability density functions given by

$$f_{K\delta} = \frac{1}{\sqrt{2\pi}\sigma_{Kv}} \exp\left(-\frac{K_v^2}{2\sigma_{Kv}^2}\right),\ f_{Kv} = \frac{1}{\sqrt{2\pi}\sigma_{K\alpha}} \exp\left(-\frac{K_\alpha^2}{2\sigma_{K\alpha}^2}\right),\ f_{K\alpha} = \frac{1}{\sqrt{2\pi}\sigma_{K\delta}} \exp\left(-\frac{K_\delta^2}{2\sigma_{K\delta}^2}\right) \quad (1)$$

Where $K_v = (v - \overline{v}) / M_v$, $K_\alpha = (\alpha - \overline{\alpha}) / M_\alpha$, $K_\delta = (\delta - \overline{\delta}) / M_\delta$ and σ_{Kv} , $\sigma_{K\alpha}$, $\sigma_{K\delta}$ are the non-dimensional standard deviations in the v , α and δ directions.

Toyoshima et al. [7] measured release velocity, release angle and their standard deviations for 2.425, 4.225 and 6.025 m shots of skilled and unskilled players. The success rates for skilled players were 74.2, 66.7 and 37.5% for these three shot ranges, respectively. The measured standard deviations in release velocity and angle were 0.18 m/s and 2.73° , 0.11 m/s, 1.04° and 0.15 m/s and 0.51° for the short, medium and long-range shots, respectively. For unskilled players, the success rates and standard deviations in release velocity and angle were 40.0%, 0.15 m/s, 1.09° ; 15.0%, 0.23 m/s, 1.06° ; and 8.3%, 0.21 m/s, 2.44° , respectively. Referring to this measured data for the three different range shots, we apply them to our short-, medium- and long-range shots and estimate the standard deviation of lateral deviation angle. For example, short-range shots of skilled and unskilled players have the ratio set of non-dimensional standard deviations of $\sigma_{Kv} : \sigma_{K\alpha} : \sigma_{K\delta} = 0.98 : 0.20 : 0.03$ and $0.75 : 0.07 : 0.66$, respectively. The ratio of non-dimensional standard deviations of the skilled and unskilled players for release velocity appears to be larger than that for release angle and lateral deviation angle.

3. Numerical simulations

We calculate ball rebound probabilities for direct and bank shots. Rebound position is defined at the moment when the height of ball center equals the hoop level on the way down. Contour lines of rebound probability for field shots with floor angles of 0 and 45° for skilled and unskilled players are shown in Figs. 1 and 2, respectively. The dashed line for the direct shots shows the top view of the optimal shot path plane including the hoop center. For bank shots, the dashed line would be the top view of the optimal ball path plane if a rigid ball with no friction bounced off a no-hoop backboard with a smooth surface. High percentage regions show locations of high rebound density and probability.

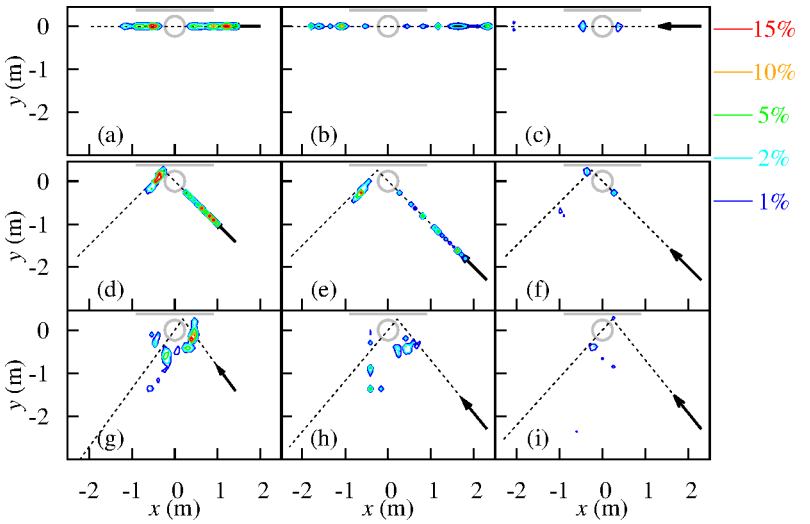


Fig. 1. Rebound probability contours for skilled players for (a) short-, (b) medium-, and (c) long-range direct shots with a floor angle of 0° ; (d) short-, (e) medium-, and (f) long-range direct shots, and (g) short-, (h) medium-, and (i) long-range bank shots with a floor angle of 45°

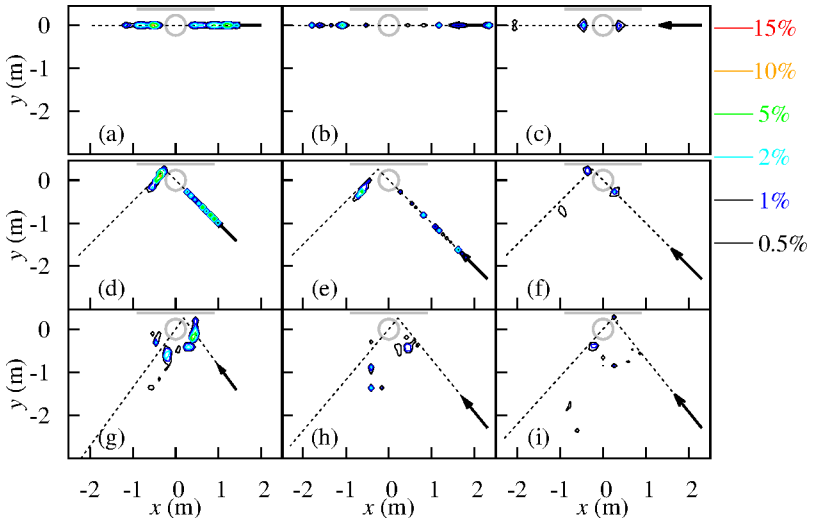


Fig. 2. Rebound probabilities for unskilled players for (a) short-, (b) medium-, and (c) long-range direct shots with a floor angle of 0° ; (d) short-, (e) medium-, and (f) long-range direct shots, and (g) short-, (h) medium-, and (i) long-range bank shots with a floor angle of 45°

Each missed shot has high rebound probability regions on the same side as and on the opposite side from the shooter. Angled direct and bank shots have different rebounding positions. Most likely rebound regions are in or near the optimal ball path plane. Rebound positions are similar in the shots of skilled and unskilled players. Maximum rebound probabilities of skilled players are higher than those of unskilled players. Okubo and Hubbard [6] have calculated rebound positions from shots with non-dimensional standard deviations of $\sigma_{Kv} = \sigma_{K\alpha} = \sigma_{K\delta}$. They found that likely rebound positions are close to the hoop on the shooter side. For short- and medium-range shots with larger standard deviations in release velocity, the rebound distance with highest probability was not close to the hoop on the shooter side. A larger standard deviation of release velocity may make the likely rebound distance longer in short- and medium-range shots.

Fig. 3 shows the high-probability rebound distance and angle from the *X* axis as a function of floor angle for short-, medium-, and long-range direct and bank shots of skilled players. There is little difference between the rebound positions of shots of skilled and unskilled players. Differences in the standard deviations of the release conditions of the skilled and unskilled players do not change the rebound position with highest probability. On the shooter side, the likely rebound position is near the hoop for bank shots. Long-range direct shots have a likely rebound position close to the hoop, but the rebound distance is apparently proportional to shot distance for short- and medium-range direct shots. Although some previous studies [4] have pointed out that missed long range shots tended to rebound far from the hoop, this conclusion is different from our results on the shooter side. On the opposite side from the shooter, the direct shot has a likely rebound position close to the hoop, and the likely rebound distances of bank shots increase as the floor angle increases. The rebound distance for long-range bank shots is apparently longer than that of short- and medium-range bank shots. Rebound angle is a function of floor angle for direct shots on the shooter side and bank shots on the opposite side, regardless of shot distance.

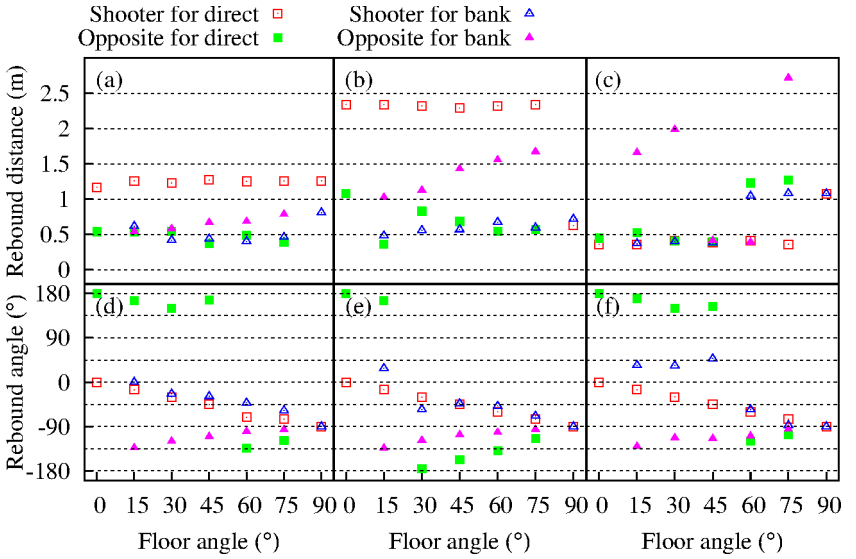


Fig. 3. Highest probability rebound distance from the hoop for (a) short-, (b) medium-, and (c) long-range shots; and angle from the *X* axis for (d) short-, (e) medium-, and (f) long-range shots of skilled players

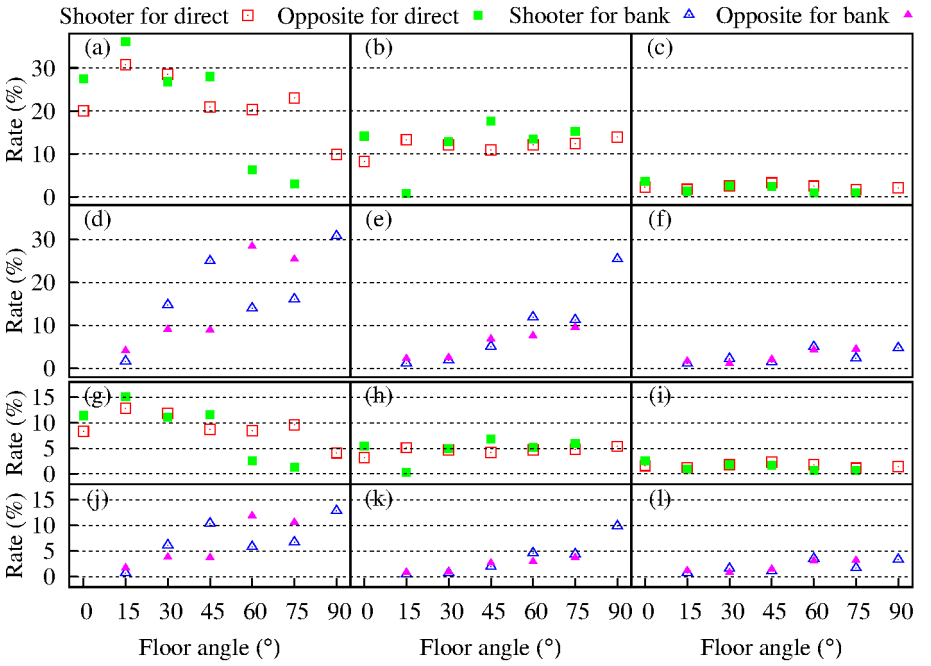


Fig. 4. Highest rebound probability of skilled players as a function of a floor angle on the same and opposite side from the shooter for (a) short-, (b) medium-, and (c) long-range direct shots; and (d) short-, (e) medium-, and (f) long-range bank shots; and (g) short-, (h) medium-, and (i) long-range direct shots, and (j) short-, (k) medium-, and (l) long-range bank shots of unskilled players.

Fig. 4 shows the rebound probabilities as a function of the floor angle on the same and opposite sides as the shooter for skilled and unskilled players. Rebound percentages of skilled players' shot are higher than those of unskilled ones. Rebound positions of skilled players' shots are more predictable than those of unskilled ones. The rebounding ball of the unskilled players can wind up more uniformly distributed around the hoop since their shots have larger standard deviations of release conditions. Some previous researchers [1-4] discussed which side of the hoop has an advantage to collect the rebounding ball. Some of these pointed out that the opposite side has an advantage for ball rebound position and others that there is no difference between the sides. In our results, it is difficult to find any comparative advantage of either side for rebounds. This may be because we assume that the aiming point is optimal. Thus it must be the center for capture and balls may equally likely bounce to both sides.

This paper focuses purely on the behavior of the ball and where it goes in a probabilistic sense after unsuccessful shots. The related problem of where a rebounding player of a given arm length should be positioned to collect a maximum number of rebounding balls is a different one. Its solution relies on the data described herein about where the ball bounces. But a player's best rebounding strategy should be to maximize his collection probability through choice of the center of circle of arm-length radius that contains the maximum summed ball location probability. We will return to this problem in a future paper.

4. Conclusions

We have calculated ball rebound positions for short-, medium-, and long-range direct and bank shots using a dynamic model and with release conditions from independent random Gaussian probability density functions. We use measured release standard deviations and assume shooters choose optimal aiming points. The relationships between the release conditions and most likely ball rebound positions has been analyzed for short-, medium- and long-range field shots. Based on these results we summarize that 1) each missed shot usually has high-probability rebound positions, one on the same as and another on the opposite side from the shooter, 2) direct and bank shots have different rebounding positions, 3) likely rebound positions usually are in or near the optimal ball path plane including the hoop center, 4) likely rebound distance from the hoop does not always depend on the distance of the shot, 5) rebounding position is more predictable with highly skilled shooters and high percentage successful shots, and 6) there is almost no advantage of either the shooter or the opposite side for highest rebound probability.

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